

On Vitalism*

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It is more than a figurative expression when one speaks of thoughts living and dying. Like all organic beings, thoughts have their determinate conditions, on which their arising, their development or their disappearance depends. Thoughts must, like organisms, struggle in order to live. Only those thoughts live in the long run which can subsist with, and be the expression of, large circles of well-tested and coherent experience. And just as a natural selection takes place among the organic germs, tentative beginnings and variations which are present at every point of development, so too a choice is constantly made among the thoughts and points of view that are available at any given time. There is a reciprocal relation between thinking and experience. For the involuntarily formed thoughts and points of view among which experience chooses have of course not themselves arisen without the influence of experience. But just as in nature far more germs and variations arise than are selected for continued development, so thinking too begins with energetic and luxuriant sallies and anticipations, which are gradually sifted, limited, developed or discarded.

This holds especially as regards the understanding of organic life. The struggle between speculation and verification moves in rhythmic fashion through the history of biology, and in the most recent period a swell seems to be astir which is leading to the taking up again of points of view whose time seemed decidedly to be past. Life has been thought to be found in old thoughts whose decease had already been announced. In what sense that opinion is justified, we shall examine more closely.

There are two fundamental conceptions which here stand over against one another. The one rests on this, that the clearest understanding we can have of a phenomenon is the one we have where we can resolve it into its elements and see it again as the product

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or result of those elements' co-operation — that is, when we go from the parts to the whole. The so-called mechanical view of nature maintains that this mode of explanation holds not only for the physical and chemical phenomena but for all natural phenomena. Against this it is urged from the other side that what is peculiar to the organic phenomena is that the constitution, the form and the mode of operation of the elements, of the parts, cannot be understood except by taking as one's ground the thought of the whole which they form, or are to form. The organism cannot be explained as a mere product of organs or cells, nor the cell as a mere product of molecules. We must then here assume a peculiar force, operating according to other laws than the mechanical ones. This is the vitalistic conception. — And since the only way in which we can represent a whole as determining the constitution and the co-operation of the parts is to represent it as a purpose that is striven after, a vitalistic conception will always remain more or less teleological.

When history shows us a rhythmic alternation of these opposed ways of seeing, the advance of knowledge is by no means thereby excluded. It is the new experiences that let the phenomena appear now more from the one side, now more from the other, and that also gradually transform the earlier forms of the two fundamental conceptions.

A mechanical view of nature first comes forward at a relatively advanced standpoint. Primitive man's conception is vitalistic or animistic, explaining all phenomena that awaken attention by the intervention of personal beings. All later vitalism has a certain kinship with this primitive conception. — In Greek antiquity the standpoint of mechanism is urged by the *Atomists* (whose doctrine was later set forth in Lucretius' famous poem *de rerum natura*). Aristotle, who was himself a brilliant investigator of nature in the domain of organic life, maintained, in opposition to atomism, a conception which regarded nature as a great scale of stages, where every member is matter and means in relation to a higher one, while it is form and purpose in relation to a lower. Nowhere in nature could he conceive the whole as arising as the result of the elements, the form as the effect of the matter. He therefore extends vitalism to the whole of nature. It is thus characteristic of the position of the problem of life in antiquity that no sharp boundary was set between inorganic and organic phenomena. For the Aristotelian conception as for the atomistic there are in nature only differences of degree.

For the time being the Aristotelian conception was victorious, and it became the prevailing one in the Middle Ages. A turning point comes with *the founding of modern natural science* by Leonardo da Vinci, Kepler and Galilei. The proposition was now set up that all change is motion, and it was demanded that this be carried through in every domain, so that all laws of nature might be derived from the principles of mechanics.

Cartesian philosophy generalized this conception and carried it out into wide circles. The discovery of the circulation of the blood and of reflex movement was an empirical support for the conviction that the domain of organic life could form no exception.

As a reaction against this whole tendency there arises the conception one now chiefly thinks of when one speaks of vitalism. *G. E. Stahl* combated mechanical physiology and maintained (especially in *Theoria medica vera* 1708) that without a soul's intervention organic growth and the organism's self-preservation were inexplicable. He thus goes further back than to Aristotle, in that he renews the primitive animism. He held that since the conscious soul can be the cause of voluntary movements, it must also be able to be the cause of the involuntary processes in the organism, even if it here operates in a more instinctive manner, and even if it does not always behave rationally — which, after all, it does not always do in its voluntary actions either. — The well-known medical school at Montpellier modified the theory by distinguishing between soul and life, so that between soul and body a peculiar life-principle was inserted. In this form vitalism is presented by *Barthez* (*Nouveaux Elements de la science de l'homme*. Montpellier 1778). He combats two sects, the Mechanists and the Animists, and represents vitalism in the narrowest sense of the word. He defines the life-principle as “the cause which brings forth all the phenomena of life in the human body”. It follows laws that lie above the laws of physics (*d'un ordre transscendant par rapport aux lois de la Physique*). Its proper nature, and especially its relation to the soul proper, is unknown. But Barthez plainly enough conceives it on the analogy of mental life proper; he ascribes to it instinct, temperament and sensory capacity. It intervenes in the life-process and directs it in a determinate direction. It lays in the muscles the force needed to overcome a determinate resistance, and shows its power especially in convulsive attacks. It dilates the muscle fibres in the ducts of milk and of semen, and it acts upon the humours, so that the substances are formed which are needed for repair or for reproduction. It regulates respiration and the development of heat, and conditions a “sympathetic” connection between the organism's different parts. — Different both from Barthez's vitalism and from Stahl's is the theory which the famous anatomist *Bichat* set up (in *Recherches physiologiques sur la vie et la mort*. Paris 1800). He conceives the vital force as a property of the organic tissues. It was an incipient decentralization of vitalism.

Classical vitalism — so I designate the conception represented, in different shadings, by Stahl, Barthez and Bichat — has this in common with the Aristotelian, that by assuming a vital force it takes itself to have given a sufficient explanation of the peculiarity of the phenomena of life. It differs from Aristotelian vitalism in sharply emphasizing the

opposition between the inorganic and the organic, and even assuming a hostile relation between them. The soul, or the vital force, must struggle with the mechanical and chemical conditions in order to get them to operate in a determinate direction; indeed Bichat even defines life as the sum of the forces that struggle against death.

A new and magnificent rise of the mechanical view of nature in its application to biology came about on the basis of the work of Italian, French and Swiss investigators of nature at the close of the eighteenth century — work which, however, only came to be properly recognized towards the middle of our own century. About the year 1840 *Dumas* and *Liebig* reformed plant physiology and demonstrated the circulation of matter in nature, and with it the connection between organic and inorganic nature. Soon after, *Robert Mayer* discovered the law of the conservation of energy and applied it at once to the organic domain (in his work *Die organische Bewegung* 1845). A proposition from that work may stand as the motto of the whole new biological conception: “In the process of life there takes place only a transformation of matter and of force, but never a creation of the one or the other.” A whole series of eminent physiologists, German ones above all (notably the disciples of *Johannes Müller* — though the master himself was still a vitalist), showed the fruitfulness of the new points of view. Physiology now seemed able to be conceived as applied physics and chemistry. The assumption of a vital force was stamped as mysticism and superstition. *Helmholtz* uses (in his work *Das Denken in der Medicin* 1878) vitalism as the opposite of natural inquiry, and regards the question of the vital force as one with the question of the perpetuum mobile. And in his memorial address on *Helmholtz* (1897) *Dubois Reymond* declared that the doctrine of the conservation of energy gave the phantom of a vital force its final blow. The period in biology which began about 1840 is one of the most fruitful in its whole history, especially through its many applications of physical and chemical laws to the organic domain. A great and coherent view of nature might hereby seem to have been won once for all.

In the latest years one has begun to speak again of vitalism as a necessary standpoint. Connected with this in a certain way is the fact that one speaks of natural science having disappointed the expectations that had been placed in it — indeed, that one has even spoken of “the bankruptcy of science”. The facts on which so-called *neo-vitalism* rests were in the main already known earlier, and were emphasized by critical investigators. Men like *Lotze* and *Virchow*, *Littre* and *Claude Bernard*, who all maintained that only the methods of physics and chemistry could lead to insight into the laws of vital phenomena, nevertheless by no means held that life’s innermost nature could be fully understood from those laws alone. They regarded life as a fact, and saw it as biology’s task to

investigate the single members of the life-process as it actually runs. The occasion of neo-vitalism's appearance can be sought only in this, that many parts of the life-process have shown themselves to be *still* more intricate than had been thought beforehand. There are three points especially that come into the foreground for neo-vitalism.

1) Even if at every single transition in the life-process there occurred only such transformations of matter and force as could be explained by chemical and physical laws, there would still remain the fact that the chemical and physical processes in growth and nutrition run in determinate *directions*. This is all the more striking in that growth takes place sporadically, from different points, so that the joining into a form of the whole comes about only gradually. Should it not then be justified to speak of a formative drive (*nisus formativus*)?

2) In *the taking up of nutriment* into the cells a choice is exercised. Neither the quantity nor the quality of what is taken up can be explained in a purely chemical or physical way. The cells seem here to act as little beings whose need and wish determine their relation to the surrounding world. Several neo-vitalists therefore personify the cells, as the classical vitalists personified the whole organism. Since we know only from ourselves what need and wish are, one of the most zealous neo-vitalists, *Bunge*, has even declared that psychology ought properly, here, to come to physiology's aid.

3) Organic tissue differs from the crystal in that its molecules *never* come into *complete equilibrium* so long as life lasts. To be able to die is therefore a mark of the living. —

It is beyond doubt that attention to these points has been sharpened in recent years. It is doubtful, on the other hand, whether a new stage has really thereby emerged in the old rhythmic process, such that the name neo-vitalism could be justified. How far one will extend the name may in any case be doubtful. Investigators like *Giulio Fano* (*La fisiologia come scienza autonoma*. Genova 1884) and *Hans Driesch* (*Die Biologie als selbständige Grundwissenschaft*. Leipzig 1893), who wish only to maintain biology's independence over against chemistry and physics, can be called neo-vitalists only improperly. The name does fit *G. Bunge*, on the other hand (the chapter "Vitalismus und Mechanismus" in "Lehrbuch der physiologischen und pathologischen Chemie". Leipzig 1887), who almost renews Stahl's animism. From standpoints very different from one another, *Haeckel*, *Verworn* and *Rindfleisch* have expressed themselves in a similar direction.¹

It will have its interest to compare neo-vitalism with classical vitalism, in order to see whether it is, as some hold (for example *Mosso* in his essay "Materialismo e Misticismo"

¹The significant difference between the conceptions that are often designated together as neo-vitalism was already pointed out by *Felice Tocco* in his essay "Le disfatte della scienza" (*Nuova Anthologia*. 1 March 1896).

in Nuova Anthologia, 1 December 1895), a sign of retrogression, or whether, in relation to earlier and more or less kindred tendencies, differences can be shown which point to advance. Such a comparison can be led back to three principal points.

1) Classical vitalism holds that with the assumption of a “vital force” a conclusive, positive explanation has been reached. But for the most critical investigators, whom neo-vitalism claims for itself, it is merely a matter of maintaining that biology has its independent empirical foundation, and is something other and more than a sum of physical and chemical deductions. In so far as a vital force is spoken of, the word means an *x*, an unresolved remainder, and stands as the expression of the intricate conditions under which the general laws of nature find application in the organic domain.

2) Classical vitalism was called by Claude Bernard “a doctrine that makes lazy” (*une doctrine paresseuse*). In its confidence in the vital force it disdained exact investigation of the particular relations. Barthez held, for example, that it was unnecessary and unreasonable to seek an anatomical basis for the connection between sensory and motor functions; the unity of the vital force was sufficient explanation of that connection. He thus rejected in advance investigations such as those which later led Charles Bell to his discovery of the difference between sensory and motor nerves. (See Barthez: *Nouveaux Elements*, p. 142–153; 207 f.)² Neo-vitalism, by contrast, holds fast — despite its inclination to psychological analogies — that the only method which can actually be in question for use in biology is the one the mechanical conception has proclaimed. It regards the problem of life as more intricate than classical vitalism did.

3) The most striking difference between neo-vitalism and classical vitalism consists in this, that the vital force is not, or not only, ascribed to the organism as a whole, but to the cells, the organism’s elements. In place of a genius directing the organism’s life as a whole, one is now inclined to assume a kind of cell-geniuses, which decide what and how much is to be taken up from the inner medium. Bunge says outright: “*Wir übertragen das aus dem eigenen Bewusstsein geschöpfte auf die Objecte unserer Sinneswahrnehmung, auf die Organe, auf die Gewebselemente, auf jede kleine Zelle.*” It undoubtedly marks an advance that the concept of force is used of the single co-operating elements instead of en bloc of the whole organism. The problem has become more localized; the riddles have been pushed further back. To be sure, the great problem then remains, how the life of the many cells is combined into the organism’s collective life — and it was this above all that awakened the wonder whose theoretical expression the old vitalism was. Neo-vitalism too is the expression of a wonder, but especially of the one awakened by

²This point was already brought out by Louis Peisse in his critique of the Montpellier school. (*La Médecine et les Medecins*. Paris 1857. I, p. 291–293).

what is mysterious in the processes that go on in the organism's smallest parts. One had not expected that the problem of life would come again at the very elements into which one thought one had resolved the whole of life. —

The difference between neo-vitalism and classical vitalism is so great that it is really unwarranted to say that it is one and the same conception that has come again. With respect to what is scientifically asserted, and to the method actually employed, no change has really taken place in relation to the immediately preceding period in the history of biology. All talk of retrogression (Mosso) or of bankruptcy (Brunetière) is therefore unwarranted, and the name neo-vitalism itself had best be avoided.

In conclusion it will be of interest to bring out the analogy between the problem which makes biology something other and more than applied chemistry and physics, and a whole series of other problems which all rest on the opposition between the rational and the empirical — or, as one may also put it, between the formal and the real in our knowledge. The more rational and formal our knowledge is, the further it can move forward through definitions and deductions, so that every transition takes place with intuitable necessity. Empirical or real knowledge, by contrast, must often stop at facts which can indeed be described and analysed, but not defined and derived from other facts. Such a fact is life; and therefore biology belongs, and will perhaps always belong, to empirical knowledge. It was not only Stahl who in his day missed a definition of what life is (a lack which he did, however, believe himself able to supply); it has been said in our own day as well (by Lotze and by Littré, by Bernard and by Fano) that no such definition has been found. But life does not stand alone in this respect. At a whole series of points the rational continuity in our knowledge is replaced by empirical data which for the time being stand as unresolvable. They are kinks in the thread, or perhaps even knots.

1) Let us suppose that the proposition that all change is motion could be completely carried through, and that the ideal of the mechanical view of nature could so far be realized. The empirical kinks or knots in the thread would not on that account be lacking. At every point — however far back we go — the motion would still appear with a certain given direction and a certain given velocity, and the outer causes which alter that direction and that velocity would operate from certain determinate places and in certain determinate degrees and directions. — The sum of energy expended in the motion, and in the causes that alter it, is moreover of a determinate given magnitude, and its distribution among the single operative elements can likewise be known only empirically.

2) Energy appears, further, in qualitatively different forms, and the operative elements likewise have their qualitative peculiarities. Even if, by the law of the conservation of energy and of matter, equivalents can always be shown at every transformation from one form of energy into another and at every change of matter, there arise at the same time in these transformations new properties which cannot be derived from the antecedent conditions, but are attached to them purely empirically.

3) In the organic domain the sum of empirical elements becomes greater still, and more pronounced. It is attention to these elements that often brings with it a renewed inclination towards vitalistic views. As such elements we have in the foregoing mentioned direction of growth and structure, qualitative and quantitative change in the cells' change of matter, and the maintenance of the unstable equilibrium.

4) If we take the psychological domain along with the rest, the phenomena of consciousness — from the most elementary sensation to the highest thoughts, feelings and acts of will — stand as an empirical given which cannot be derived from the material presuppositions. Psychology is to a still higher degree than the other parts of biology a purely descriptive science. And within psychology there then come forward again new qualitative differences and transformations, where the same problem presents itself anew.

It is clear that if, at all the points where the empirical, the qualitative and the individual put to the test the rational continuity our knowledge strives after, one were to assume new independent and original "forces", one would be deceiving oneself. The limits we thus constantly come up against are always at the same time to be regarded as new material, setting new tasks. It is existence's richness and depth that produces these kinks and knots in the thread, and thereby makes necessary a constant interaction between experience and thinking, induction and deduction, analysis and synthesis — that doubleness in method which Goethe compared with the relation between breathing in and breathing out. The struggle between mechanism and "vitalism" is, rightly understood, a struggle between two tendencies between which there can at bottom be no opposition. We should not be more perfect beings if we could breathe out without breathing in. That doubleness makes both life and thought the richer.